제어공학 문제 풀이 과제

2024522기 이찬호

8-1. 다음 미분방정식은 선형시불변시스템을 나타낸다. 동적방정식(상태방정식 및 출력방정식)을 벡터-행렬 형식으로 나타내어라.

(a) $\frac{d^2y(t)}{dt^2}+4\frac{dy(t)}{dt}+y(t)=5r(t)$

(b) $2\frac{d^3y(t)}{dt^3}+3\frac{d^2y(t)}{dt^2}+5\frac{dy(t)}{dt}+2y(t)=r(t)$

(a) $\ddot{y}+4\dot{y}+y=5r$

$x_1=y,\ x_2=\dot{y},\ \dot{x}_1=\dot{y}=x_2,\ \ddot{y}=\dot{x}_2$

$\dot{x}_2+4x_2+x_1=5r$

$\dot{x}_2=-x_1-4x_2+5r$

$\dot{x}_1=x_2$

상태방정식 $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -1 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 5\end{bmatrix}r$

출력방정식 $y=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$

(b) $2\dddot{y}+3\ddot{y}+5\dot{y}+2y=r$

$x_1=y,\ x_2=\dot{y},\ \dot{x}_1=\dot{y}=x_2,\ x_3=\ddot{y},\ \dot{x}_2=\ddot{y}=x_3,\ \dddot{y}=\dot{x}_3$

$2\dot{x}_3+3x_3+5x_2+2x_1=r$

$\dot{x}_3=-x_1-\frac{5}{2}x_2-\frac{3}{2}x_3+\frac{1}{2}r$

상태방정식 $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -\frac{5}{2} & -\frac{3}{2}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ \frac{1}{2}\end{bmatrix}r$

출력방정식 $y=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$

8-2. 다음의 전달함수들은 선형시불변시스템을 나타낸다. 동적방정식(상태방정식 및 출력방정식)을 벡터-행렬 형식으로 나타내어라.

(a) $G(s)=\frac{s+3}{s^2+3s+2}$ (c) $G(s)=\frac{s+2}{s^2+7s+12}$

(b) $G(s)=\frac{6}{s^3+6s^2+11s+6}$ (d) $G(s)=\frac{s^3+11s^2+35s+250}{s^3(s^3+4s^2+39s+108)}$

(a) $G(s)=\frac{s+3}{s^2+3s+2}=\frac{Y(s)}{U(s)}$

$\frac{X(s)}{U(s)}=\frac{1}{s^2+3s+2} \Rightarrow s^2X(s)+3sX(s)+2X(s)=U(s)$

$\Rightarrow \ddot{x}x+3\dot{x}+2x=u \quad (x_1=x,\ x_2=\dot{x},\ \dot{x}_1=\dot{x}=x_2)$

$\Rightarrow \dot{x}_2+3x_2+2x_1=u$

$\Rightarrow \dot{x}_2=-2x_1-3x_2+u$

상태방정식 $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}+\begin{bmatrix}0\\ 1\end{bmatrix}u$

$\frac{Y(s)}{X(s)}=s+3 \Rightarrow Y(s)=sX(s)+3X(s)$

$=\dot{x}+3x$

$=x_2+3x_1$

출력방정식 $y=\begin{bmatrix}3 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$

(b) $G(s)=\frac{6}{s^3+6s^2+11s+6}=\frac{Y(s)}{U(s)}$

$\frac{X(s)}{U(s)}=\frac{1}{s^3+6s^2+11s+6} \Rightarrow s^3X(s)+6s^2X(s)+11sX(s)+6X(s)=U(s)$

$\Rightarrow \dddot{x}+6\ddot{x}+11\dot{x}+6x=u \quad (x_1=x,\ x_2=\dot{x},\ x_3=\ddot{x},\ \dot{x}_1=\dot{x}=x_2,\ \dot{x}_2=\ddot{x}=x_3)$

$\Rightarrow \dot{x}_3+6x_3+11x_2+6x_1=u$

$\Rightarrow \dot{x}_3=-6x_1-11x_2-6x_3+u$

상태방정식 $\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -6 & -11 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$

$\frac{Y(s)}{X(s)}=6$ $Y(s)=6X(s)$

$=6x$

$=6x_1$

출력방정식 $y=\begin{bmatrix}6 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$

8-13. 다음과 같은 동적방정식이 주어졌다.

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad y(t) = \mathbf{C}\mathbf{x}(t)$$

(a) $\mathbf{A} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$; $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$; $\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 1 & 1 \end{bmatrix}$; $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$; $\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -2 & 0 \\ -1 & -2 & -3 \end{bmatrix}$; $\mathbf{B} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$; $\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$

(d) $\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$; $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$; $\mathbf{C} = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$

(e) $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ -2 & -3 \end{bmatrix}$; $\mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$; $\mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}$

상태방정식을 가제어성표준형(CCF)으로 변환하는 변환 $\mathbf{x}(t) = \mathbf{P}\bar{\mathbf{x}}(t)$를 구하라.

(a) $A = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $SI - A = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s & -2 & 0 \\ -1 & s+2 & 0 \\ 1 & 0 & s-1 \end{bmatrix}$

$$|SI - A| = 0 - 0 + (s-1)\begin{vmatrix} s & -2 \\ -1 & s-2 \end{vmatrix}$$

$$= (s-1)\{s(s-2) - (-2-1)\}$$

$$= (s-1)(s^2 - 2s - 2) = s^3 - 3s^2 + 2 \Longleftrightarrow s^3 + a_2 s^2 + a_1 s + a_0$$

$a_0 = 2$, $a_1 = 0$, $a_2 = -3$

$$\bar{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 0 & 3 \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$M = [B \;\; AB \;\; A^2B]$

$$AB = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad A^2B = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -1 \end{bmatrix}$$

$\therefore$, $M = \begin{bmatrix} 0 & 2 & 4 \\ 1 & 2 & 6 \\ 1 & 1 & -1 \end{bmatrix}$

$\bar{M} = [\bar{B} \;\; \bar{A}\bar{B} \;\; \bar{A}^2\bar{B}]$

$$\bar{A}\bar{B} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 0 & 3 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \quad \bar{A}^2\bar{B} = \bar{A}\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}$$

$\therefore$, $\bar{M} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 3 \\ 1 & 3 & 9 \end{bmatrix}$

$P = M\bar{M}^{-1}$ $\bar{M}^{-1} = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & -3 & 1 \\ 0 & 1 & 0 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$

$$P = \begin{bmatrix} 0 & 2 & 4 \\ 1 & 2 & 6 \\ 1 & 1 & -1 \end{bmatrix}\begin{bmatrix} 0 & -3 & 1 \\ -3 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -2 & 2 & 0 \\ 0 & -1 & 1 \\ -4 & -2 & 1 \end{bmatrix}$$

$\therefore$, $x(t) = \begin{bmatrix} -2 & 2 & 0 \\ 0 & -1 & 1 \\ -4 & -2 & 1 \end{bmatrix}\bar{x}(t)$

(c)

$$A=\begin{bmatrix}-2&1&0\\0&-2&0\\-1&-2&-3\end{bmatrix} \quad B=\begin{bmatrix}1\\1\\1\end{bmatrix} \quad SI-A=\begin{bmatrix}s+2&-1&0\\0&s+2&0\\1&2&s+3\end{bmatrix}$$

$$|SI-A|=(s+3)\begin{vmatrix}s+2&-1\\0&s+2\end{vmatrix}=(s+3)(s+2)^2=(s+3)(s^2+4s+4)$$

$$=s^3+4s^2+4s+3s^2+12s+12=s^3+7s^2+16s+12$$

$$a_0=12 \quad a_1=16 \quad a_2=7$$

$$\bar{A}=\begin{bmatrix}0&1&0\\0&0&1\\-12&-16&-7\end{bmatrix} \quad \bar{B}=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$M=[B \ AB \ A^2B]$$

$$AB=\begin{bmatrix}-2&1&0\\0&-2&0\\-1&-2&-3\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}-1\\-2\\-6\end{bmatrix} \quad A^2B=\begin{bmatrix}-2&1&0\\0&-2&0\\-1&-3&-3\end{bmatrix}\begin{bmatrix}-1\\-2\\6\end{bmatrix}=\begin{bmatrix}0\\4\\23\end{bmatrix} \quad M=\begin{bmatrix}1&-1&0\\1&-2&4\\1&6&23\end{bmatrix}$$

$$\bar{M}=[\bar{B} \ \bar{A}\bar{B} \ \bar{A}^2\bar{B}]$$

$$\bar{A}\bar{B}=\begin{bmatrix}0&1&0\\0&0&1\\-12&-16&-7\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}0\\1\\-7\end{bmatrix} \quad \bar{A}^2\bar{B}=\begin{bmatrix}0&1&0\\0&0&1\\-12&-16&-7\end{bmatrix}\begin{bmatrix}0\\1\\-7\end{bmatrix}=\begin{bmatrix}1\\-7\\33\end{bmatrix} \quad \bar{M}=\begin{bmatrix}0&0&1\\0&1&-7\\1&-7&33\end{bmatrix}$$

$$\bar{M}^{-1}=\begin{bmatrix}0&0&1&1&0&0\\0&1&-7&0&1&0\\1&-7&33&0&0&1\end{bmatrix}=\begin{bmatrix}1&-7&33&0&0&1\\0&1&-7&0&1&0\\0&0&1&1&0&0\end{bmatrix}=\begin{bmatrix}1&0&0&16&7&1\\0&1&0&7&1&0\\0&0&1&1&0&0\end{bmatrix}$$

$$P=\begin{bmatrix}1&-1&0\\1&-2&4\\1&-6&23\end{bmatrix}\begin{bmatrix}16&7&1\\7&1&0\\1&0&0\end{bmatrix}=\begin{bmatrix}9&6&1\\6&5&1\\-3&1&1\end{bmatrix}$$

$$x(t)=\begin{bmatrix}9&6&1\\6&5&1\\-3&1&1\end{bmatrix}\bar{x}(t)$$

(e)

$A=\begin{bmatrix}1 & 1\\ -2 & -3\end{bmatrix}$ $B=\begin{bmatrix}0\\ 1\end{bmatrix}$ $SI-A=\begin{bmatrix}S-1 & -1\\ 2 & S+3\end{bmatrix}$ $|SI-A|=S^2+2S-1$

$a_0=-1, a_1=2$

$\bar{A}=\begin{bmatrix}0 & 1\\ -a_0 & -a_1\end{bmatrix}=\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}$, $\bar{B}=\begin{bmatrix}0\\ 1\end{bmatrix}$

$M=[B\ AB]$

$AB=\begin{bmatrix}1 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}=\begin{bmatrix}1\\ -3\end{bmatrix}$ $M=\begin{bmatrix}0 & 1\\ 1 & -3\end{bmatrix}$

$\bar{M}=[\bar{B}\ \bar{A}\bar{B}]$

$\bar{A}\bar{B}=\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}=\begin{bmatrix}1\\ -2\end{bmatrix}$ $\bar{M}=\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}$

$\bar{M}^{-1}=\begin{bmatrix}0 & 1 & 1 & 0\\ 1 & -2 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & -2 & 0 & 1\\ 0 & 1 & 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0 & 2 & 1\\ 0 & 1 & 1 & 0\end{bmatrix}$

$P=\begin{bmatrix}0 & 1\\ 1 & -3\end{bmatrix}\begin{bmatrix}2 & 1\\ 1 & 0\end{bmatrix}=\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}$

$x(t)=\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}\bar{x}(t)$

(a) $A=\begin{bmatrix}0&2&0\\1&2&0\\-1&0&1\end{bmatrix}$; $B=\begin{bmatrix}0\\1\\1\end{bmatrix}$; $C=\begin{bmatrix}1&0&1\end{bmatrix}$

(b) $A=\begin{bmatrix}0&2&0\\1&2&0\\-1&1&1\end{bmatrix}$; $B=\begin{bmatrix}1\\1\\0\end{bmatrix}$; $C=\begin{bmatrix}1&0&1\end{bmatrix}$

(c) $A=\begin{bmatrix}-2&1&0\\0&-2&0\\-1&-2&-3\end{bmatrix}$; $B=\begin{bmatrix}1\\1\\1\end{bmatrix}$; $C=\begin{bmatrix}1&0&0\end{bmatrix}$

(d) $A=\begin{bmatrix}-1&1&0\\0&-1&1\\0&0&-1\end{bmatrix}$; $B=\begin{bmatrix}0\\1\\1\end{bmatrix}$; $C=\begin{bmatrix}1&0&1\end{bmatrix}$

(e) $A=\begin{bmatrix}1&1\\-2&-3\end{bmatrix}$; $B=\begin{bmatrix}0\\1\end{bmatrix}$; $C=\begin{bmatrix}1&0\end{bmatrix}$

8-14. 문제 8-13에 기술된 시스템에 대하여 상태방정식을 가관측성표준형(OCF)으로 변환하는 변환 $\mathbf{x}(t)=\mathbf{Q}\bar{\mathbf{x}}(t)$를 구하라.

(a)
$$A=\begin{bmatrix}0&2&0\\1&2&0\\-1&0&1\end{bmatrix}\quad C=\begin{bmatrix}1&0&1\end{bmatrix}$$

$$CA=\begin{bmatrix}1&0&1\end{bmatrix}\begin{bmatrix}0&2&0\\1&2&0\\-1&0&1\end{bmatrix}=\begin{bmatrix}-1&2&1\end{bmatrix}\qquad CA^2=\begin{bmatrix}-1&2&1\end{bmatrix}\begin{bmatrix}0&2&0\\1&2&0\\-1&0&1\end{bmatrix}=\begin{bmatrix}1&2&1\end{bmatrix}$$

$$N=\begin{bmatrix}C\\CA\\CA^2\end{bmatrix}=\begin{bmatrix}1&0&1\\-1&2&1\\1&2&1\end{bmatrix}$$

$$SI-A=\begin{bmatrix}S&-2&0\\-1&S-2&0\\1&0&S-1\end{bmatrix}\qquad |SI-A|=(S-1)(S^2-2S-2)=S^3-3S^2+2$$

$$a_2=-3,\ a_1=0,\ a_0=2$$

$$\bar{A}=\begin{bmatrix}0&0&-a_0\\1&0&-a_1\\0&1&-a_2\end{bmatrix}=\begin{bmatrix}0&0&-2\\1&0&0\\0&1&3\end{bmatrix}\qquad \bar{C}=\begin{bmatrix}0&0&1\end{bmatrix}$$

$$\bar{C}\bar{A}=\begin{bmatrix}0&0&1\end{bmatrix}\begin{bmatrix}0&0&-2\\1&0&0\\0&1&3\end{bmatrix}=\begin{bmatrix}0&1&3\end{bmatrix}\qquad \bar{C}\bar{A}^2=\begin{bmatrix}0&1&3\end{bmatrix}\begin{bmatrix}0&0&-2\\1&0&0\\0&1&3\end{bmatrix}=\begin{bmatrix}1&3&9\end{bmatrix}$$

$$\bar{N}=\begin{bmatrix}0&0&1\\0&1&3\\1&3&9\end{bmatrix}\qquad N^{-1}=\begin{bmatrix}1&0&1&1&0&0\\-1&2&1&0&1&0\\1&2&1&0&0&1\end{bmatrix}=\begin{bmatrix}1&0&0&0&-\frac{1}{2}&\frac{1}{2}\\0&1&0&-\frac{1}{2}&0&\frac{1}{2}\\0&0&1&1&\frac{1}{2}&-\frac{1}{2}\end{bmatrix}$$

$$Q=\begin{bmatrix}0&-\frac{1}{2}&\frac{1}{2}\\-\frac{1}{2}&0&\frac{1}{2}\\1&\frac{1}{2}&-\frac{1}{2}\end{bmatrix}\begin{bmatrix}0&0&1\\0&1&3\\1&3&9\end{bmatrix}=\begin{bmatrix}\frac{1}{2}&1&3\\\frac{1}{2}&\frac{3}{2}&4\\-\frac{1}{2}&-1&-2\end{bmatrix}\qquad x(t)=\begin{bmatrix}\frac{1}{2}&1&3\\\frac{1}{2}&\frac{3}{2}&4\\-\frac{1}{2}&-1&-2\end{bmatrix}\bar{x}(t)$$

(c) $A=\begin{bmatrix}-2 & 1 & 0\\ 0 & -2 & 0\\ -1 & -2 & -3\end{bmatrix}$ $C=\begin{bmatrix}1 & 0 & 0\end{bmatrix}$

$CA=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}-2 & 1 & 0\\ 0 & -2 & 0\\ -1 & -2 & -3\end{bmatrix}=\begin{bmatrix}-2 & 1 & 0\end{bmatrix}$, $CA^2=\begin{bmatrix}-2 & 1 & 0\end{bmatrix}\begin{bmatrix}-2 & 1 & 0\\ 0 & -2 & 0\\ -1 & -2 & -3\end{bmatrix}=\begin{bmatrix}4 & -4 & 0\end{bmatrix}$

$N=\begin{bmatrix}1 & 0 & 0\\ -2 & 1 & 0\\ 4 & -4 & 0\end{bmatrix}$ $|N|=0$ 이므로 N^{-1}이 존재X 즉 Q가 존재하지 않음 (가관측성이 없다)

(e) $A=\begin{bmatrix}1 & 1\\ -2 & -3\end{bmatrix}$, $C=\begin{bmatrix}1 & 0\end{bmatrix}$ $CA=\begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ -2 & -3\end{bmatrix}=\begin{bmatrix}1 & 1\end{bmatrix}$

$N=\begin{bmatrix}C\\ CA\end{bmatrix}=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$ $|SI-A|=S^2+2S-1$ $a_1=2$, $a_0=-1$

$\bar{A}=\begin{bmatrix}0 & -a_0\\ 1 & -a_1\end{bmatrix}=\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}$, $\bar{C}=\begin{bmatrix}0 & 1\end{bmatrix}$

$\bar{C}\bar{A}=\begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}=\begin{bmatrix}1 & -2\end{bmatrix}$

$\bar{N}=\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}$ $N^{-1}=\begin{bmatrix}1 & 0 & 1 & 0\\ 1 & 1 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0 & 1 & 0\\ 0 & 1 & -1 & 1\end{bmatrix}$

$Q=N^{-1}\bar{N}=\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & -2\end{bmatrix}=\begin{bmatrix}0 & 1\\ 1 & -3\end{bmatrix}$

$\dot{x}=\begin{bmatrix}0 & 1\\ 1 & -3\end{bmatrix}\bar{x}$

8-16. 다음의 전달함수에 대하여 각각의 상태방정식을 가제어성표준형(CCF)과 가관측성표준형(OCF)으로 변환하라.

(a) $\frac{s^2-1}{s^2(s^2-2)}$ (b) $\frac{2s+1}{s^2+4s+4}$

(a) $\frac{S^2-1}{S^4-2S^2} \Leftrightarrow b_3S^3+b_2S^2+b_1S+b_0$ $b_0=-1$, $b_1=0$, $b_2=1$, $b_3=0$

$S^4+a_3S^3+a_2S^2+a_1S+a_0$ $a_0=0$, $a_1=0$, $a_2=-2$, $a_3=0$

$$A_c=\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\-a_0&-a_1&-a_2&-a_3\end{bmatrix}=\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\0&0&2&0\end{bmatrix} \quad B_c=\begin{bmatrix}0\\0\\0\\1\end{bmatrix} \quad C_c=[b_0\ b_1\ b_2\ b_3]=[-1\ 0\ 1\ 0]$$

CCF 상태방정식 $\dot{x}=\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\0&0&2&0\end{bmatrix}x+\begin{bmatrix}0\\0\\0\\1\end{bmatrix}u \quad y=[-1\ 0\ 1\ 0]x$

A_0는 CCF행렬의 행과 열을 바꾼것과 같다.

$$A_0=A_c^T=\begin{bmatrix}0&0&0&0\\1&0&0&0\\0&1&0&2\\0&0&1&0\end{bmatrix} \quad B_0=C_c^T=\begin{bmatrix}-1\\0\\1\\0\end{bmatrix} \quad C_0=B_c^T=[0\ 0\ 0\ 1]$$

OCF 상태방정식 $\dot{z}=\begin{bmatrix}0&0&0&0\\1&0&0&0\\0&1&0&2\\0&0&1&0\end{bmatrix}z+\begin{bmatrix}-1\\0\\1\\0\end{bmatrix}u \quad y=[0\ 0\ 0\ 1]z$

(b) $\frac{2s+1}{S^2+4s+4} \Leftrightarrow b_1S+b_0$ $b_0=1$, $b_1=2$

$S^2+4s+4 \Leftrightarrow S^2+a_1S+a_0$ $a_0=4$, $a_1=4$

$$A_c=\begin{bmatrix}0&1\\-a_0&-a_1\end{bmatrix}=\begin{bmatrix}0&1\\-4&-4\end{bmatrix} \quad B_c=\begin{bmatrix}0\\1\end{bmatrix} \quad C_c=[b_0,b_1]=[1\ 2]$$

CCF 상태방정식 $\dot{x}=\begin{bmatrix}0&1\\-4&-4\end{bmatrix}x+\begin{bmatrix}0\\1\end{bmatrix}u \quad y=[1\ 2]x$

$$A_0=A_c^T=\begin{bmatrix}0&-4\\1&-4\end{bmatrix}, \quad B_0=C_c^T=\begin{bmatrix}1\\2\end{bmatrix} \quad C_0=B_c^T=[0\ 1]$$

OCF 상태방정식 $\dot{z}=\begin{bmatrix}0&-4\\1&-4\end{bmatrix}z+\begin{bmatrix}1\\2\end{bmatrix}u \quad y=[0\ 1]z$

8-18. 다음 시스템들의 가제어성을 검사하라.

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$

(a) $A=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ $B=\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ $AB=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \end{bmatrix}=\begin{bmatrix} -2 \\ -10 \end{bmatrix}$

$Q=[B\ AB]=\begin{bmatrix} 2 & -2 \\ 5 & -10 \end{bmatrix}$

$|Q|=-10$, 즉 결과가 0이 아니므로 완전 가제어이다

(b) $A=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ $B=\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $AB=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \end{bmatrix}=\begin{bmatrix} -2 \\ 0 \end{bmatrix}$

$Q=[B\ AB]=\begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix}$

$|Q|=0$, 즉 결과가 0 이므로 가제어가 아니다.

8-19. 다음 시스템들의 가관측성을 검사하라.

(a) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \quad y=\begin{bmatrix} 1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(b) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \quad y=\begin{bmatrix} 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

(a) $A=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ $C=[1\ 3]$ $CA=[1\ 3]\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}=[-1\ -6]$

$Q_o=\begin{bmatrix} C \\ CA \end{bmatrix}=\begin{bmatrix} 1 & 3 \\ -1 & -6 \end{bmatrix}$ $|Q_o|=-3$, 즉 결과가 0이 아니므로 완전 가관측이다

(b) $A=\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, $C=[0\ 1]$ $CA=[0\ 1]\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}=[0\ -2]$

$Q_o=\begin{bmatrix} C \\ CA \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$ $|Q_o|=0$, 즉 결과가 0 이므로 완전 가관측이 아니다

8-47. 선형제어시스템의 전달함수가 다음과 같이 주어져 있다.

$$\frac{Y(s)}{R(s)}=\frac{s+\alpha}{s^3+7s^2+14s+8}$$

(a) 이 시스템을 제어불가능하게 하거나 관측불가능하게 하는 α 값을 정하라.
(b) (a)에서 구한 α 값으로 상태변수 중 하나가 제어불가능하게 되는 상태변수를 정의하라.
(c) (a)에서 구한 α의 값으로 상태변수 중 하나가 관측불가능하게 되는 상태변수를 정의하라.

(a) 분모인 $s^3+7s^2+14s+8=(s+1)(s^2+6s+8)=(s+1)(s+2)(s+4)$

즉, $\frac{Y(s)}{R(s)}=\frac{s+\alpha}{(s+1)(s+2)(s+4)}$

$\alpha=1, 2, 4$

(b) $\frac{s+\alpha}{(s+1)(s+2)(s+4)}=\frac{k_1}{s+1}+\frac{k_2}{s+2}+\frac{k_3}{s+4}$

$s+\alpha=k_1(s+2)(s+4)+k_2(s+1)(s+4)+k_3(s+1)(s+2)$

① $s=-1$, $k_1=\frac{\alpha-1}{3}$
② $s=-2$, $k_2=\frac{2-\alpha}{2}$
③ $s=-4$, $k_3=\frac{\alpha-4}{6}$

즉, $\frac{Y(s)}{R(s)}=\frac{\frac{(\alpha-1)}{3}}{s+1}+\frac{\frac{(2-\alpha)}{2}}{s+2}+\frac{\frac{(\alpha-4)}{6}}{s+4}$

$\dot{x}_1=-x_1+\frac{\alpha-1}{3}r$
$\dot{x}_2=-2x_2+\frac{2-\alpha}{2}r$
$\dot{x}_3=-4x_3+\frac{\alpha-4}{6}r$
$y=x_1+x_2+x_3$

α	입력계수 0행렬	제어 불능 상태
1	$k_1=0$	x_1
2	$k_2=0$	x_2
3	$k_3=0$	x_3

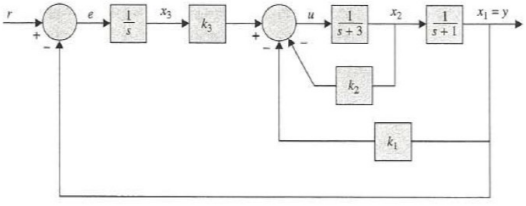
(c) $\dot{x}_1=-x_1+r$
$\dot{x}_2=-2x_2+r$
$\dot{x}_3=-4x_3+r$
$y=\frac{\alpha-1}{3}x_1+\frac{2-\alpha}{2}x_2+\frac{\alpha-4}{6}x_3$

α	출력 계수 0행렬	관측 불능 상태
1	$k_1=0$	x_1
2	$k_2=0$	x_2
3	$k_3=0$	x_3

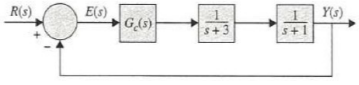
8-59. 상태피드백을 사용한 제어시스템의 블록선도를 그림 8P-59*a*에 도시하였다. 피드백 이득 k_1, k_2 및 k_3는 실수이다.

(a) 다음을 만족시키는 피드백 이득을 구하라.

- 계단입력에 대한 정상상태오차 e_{ss}[$e(t)$는 오차신호]는 영이다.
- 특성방정식의 복소근은 $-1+j$와 $-1-j$ 및 -10에 있다.



(a)



(b)

그림 8P-59

(b) 상태피드백을 사용하는 대신 그림 8P-59*b*에 도시된 직렬 연결된 제어기가 구현되어 있다. (a)에서 구한 k_1, k_2 및 k_3와 기타 시스템 파라미터를 이용하여 제어기 $G_c(s)$의 전달함수를 구하라.

(a) $e = r - y$ $\quad X_3(s) = \frac{E(s)}{s}$

대입

$u = k_3x_3 - k_2x_2 - k_1x_1$, $x_1 = y$ $\quad X_2 = \frac{U}{s+3}$, $Y = \frac{X_2}{s+1}$ $\quad X_2 = (s+1)Y$, $U = (s+3)X_2$ $\Rightarrow U = (s+3)(s+1)Y$

$U = k_3X_3 - k_2X_2 - k_1Y$

$X_3 = \frac{R-Y}{s}$, $X_2 = (s+1)Y$, $U = (s+3)(s+1)Y$

$(s+3)(s+1)Y = \frac{k_3}{s}(R-Y) - k_2(s+1)Y - k_1Y$

$s(s+3)(s+1)Y = k_3R - k_3Y - k_2s(s+1)Y - k_1sY$

$$\frac{Y}{R} = \frac{k_3}{s(s+3)(s+1) + k_2s(s+1) + k_1s + k_3} = \frac{k_3}{s^3 + (4+k_2)s^2 + (3+k_2+k_1)s + k_3}$$

비교

극점 $-1+j$, $-1-j$, -10

$(s+1-j)(s+1+j)(s+10) = s^3 + 10s^2 + 2s^2 + 20s + 2s + 20 = s^3 + 12s^2 + 22s + 20$

$k_1 = 11$, $k_2 = 8$, $k_3 = 20$

$$T(s) = \frac{Y(s)}{R(s)} = \frac{20}{s^3 + 12s^2 + 22s + 20}$$

$R(s) = \frac{1}{s}$, $E = R - Y$ $\Rightarrow E(s) = [1 - T(s)]\frac{1}{s}$

$$e_{ss} = \lim_{s \to 0} sE(s) = \lim_{s \to 0} [1 - T(s)] = 1 - \frac{20}{20} = 0 \qquad e_{ss} = 0$$

즉 $k_1 = 11$, $k_2 = 8$, $k_3 = 20$, $e_{ss} = 0$ 이다

(b) $E \to G_c(S) \to U$

$G_c(S) = \frac{U(S)}{E(S)}$

$U = 20X_3 - 8X_2 - 11Y$

$X_3 = \frac{E}{S}$, $X_2 = \frac{U}{S+3}$, $Y = \frac{U}{(S+3)(S+1)}$

$U = \frac{20}{S}E - \frac{8}{S+3}U - \frac{11}{(S+3)(S+1)}U$

$\frac{(S+3)(S+1)+8(S+1)+11}{(S+3)(S+1)}U = \frac{20}{S}E$

$U = \frac{20(S+3)(S+1)}{S(S^2+12S+22)}E$ $\quad G_c(S) = \frac{U(S)}{E(S)} = \frac{20(S+3)(S+1)}{S(S^2+12S+22)}$

즉 $G_c(S) = \frac{20(S+3)(S+1)}{S(S^2+12S+22)}$ 이다.

8-65. 다음과 같은 시스템의 상태공간 방정식을 고려하자.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

(a) 다음과 같은 사양이 만족되도록 상태피드백 제어기를 설계하라.

(i) 감쇠계수 $\zeta = 0.707$

(ii) 단위계단응답에서 마루시간(peak time)은 3초

$U = -k_1x_1 - k_2x_2 = Kx$, $K = [k_1\ k_2]$

$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$, $3 = \frac{\pi}{\omega_n\sqrt{1-0.707^2}}$, $3\omega_n\sqrt{1-0.707^2} = \pi$

$\omega_n = \frac{\pi}{3\sqrt{1-0.707^2}} = 1.48$ [rad/s]

$S^2 + 2\zeta\omega_n S + \omega_n^2 = 0$

$S^2 + 2.094S + 2.193 = 0$

$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -6x_1 - 5x_2 + u \end{cases}$

$A - BK = \begin{bmatrix} 0 & 1 \\ -(6+k_1) & -(5+k_2) \end{bmatrix}$

$SI - (A-BK) = \begin{bmatrix} S & -1 \\ 6+k_1 & S+5+k_2 \end{bmatrix}$ $\quad |SI-(A-BK)| = S^2 + (5+k_2)S + (6+k_1) = 0$

$\underbrace{S^2 + (5+k_2)S + (6+k_1)}_{\text{피드백 시스템}} = \underbrace{S^2 + 2.094S + 2.193}_{\text{원하는 시스템}}$

$k_1 = -3.807$ $\quad k_2 = -2.906$

$K = [-3.807\ \ -2.906]$, $U = 3.807x_1 + 2.906x_2$